

Adama Science and Technology University



School of Applied Natural Science

Department of Applied Chemistry

Model Exist Exam-II

For 4th Year Undergraduate Applied Chemistry Students

General Instructions

Model exist exam-II of Chemistry contains 100 questions. The questions consist of A-D multiple chooses and then choose the best answer from the given alternatives and provide your answer on the space provided only.

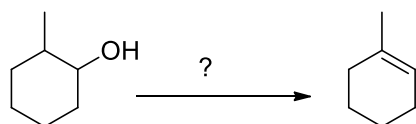
May, 2023

Adama, Ethiopia

1. As the molecular weight of alkane's increases, how do the boiling point and melting point change?

2. Which of the following compounds will undergo an S_N2 reaction most readily?

3. Which is the best reagent for carrying out the following conversion?



- A. LiAlH_4 C. H_2/Ni
B. Conc H_2SO_4 D. NaOH

4. What is the IUPAC name for the following structure? $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$

- A. 4-methyl-2-pentanol
B. 2-methyl-2-hexanol
C. 2-methyl-4-pentanol
D. 2-methyl-3-pentanol

5. Which of the following reagents can be used to oxidize 1° alcohols to aldehydes?

- A. KMnO_4
B. MnO_2
- C. $\text{K}_2\text{Cr}_2\text{O}_7$
D. PCC

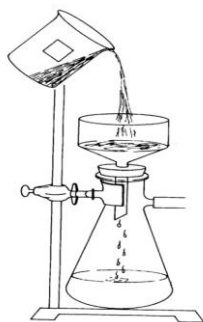
6. Acetone undergoes reduction with zinc amalgam in HCl to form propane. This reaction is known as

- A. Clemmensen reduction
B. Wolf-Kishner reduction
C. Rosenmund reduction
D. Aldol condensation

7. What condition (s) must be met for a successful liquid-liquid extraction

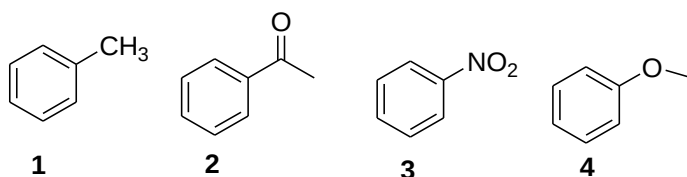
- A. The two solvents have same densities
- B. The solute is preferentially more soluble in one of the two solvents
- C. The two solvents are immiscible
- D. The two solvents have same boiling point

8. The name of the following apparatus is



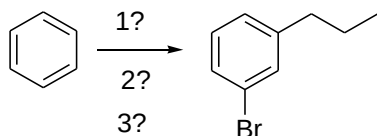
A. Gravity filtration B. Suction filtration C. Simple distillation D. separatory funnel

9. Which of the following has the compounds shown in the correct order of reactivity towards electrophilic aromatic substitution with fuming nitric acid?



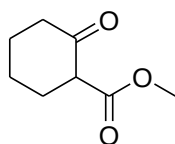
- A. 1>2>4>3
 B. 3>2>1>4
 C. 4>1>2>3
 D. 3>2>4>1

10. Which is the best reaction sequence to synthesize *m*-bromopropyl benzene from benzene?



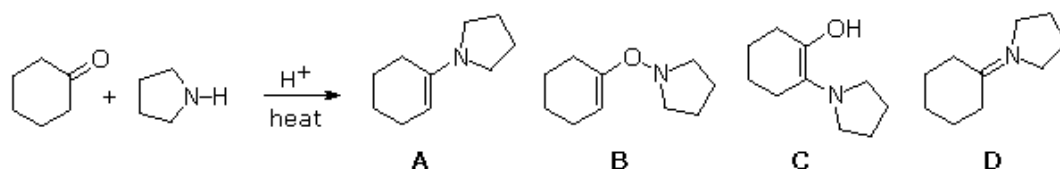
- A. 1) $\text{Br}_2/\text{FeBr}_3$, 2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}/\text{FeBr}_3$, 3) $\text{Zn}(\text{Hg})/\text{HCl}$,
 B. 1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCl}/\text{FeCl}_3$, 2) $\text{Zn}(\text{Hg})/\text{HCl}$, 3) $\text{Br}_2/\text{FeBr}_3$
 C. 1) $\text{CH}_3\text{CH}_2\text{COCl}/\text{FeCl}_3$, 2) $\text{Zn}(\text{Hg})/\text{HCl}$, 3) $\text{Br}_2/\text{FeBr}_3$
 D. 1) $\text{CH}_3\text{CH}_2\text{COCl}/\text{FeCl}_3$, 2) $\text{Br}_2/\text{FeBr}_3$, 3) $\text{Zn}(\text{Hg})/\text{HCl}$

11. The ester shown below can be formed by a Dieckmann cyclization. The starting material would be:



- A. diethyl-1,6-hexanedioate
 B. cyclohexanone
 C. diethyl-1,8-octanedioate
 D. diethyl-1,7-heptanedioate

12. Which of the following will be the favored product from the reaction shown below?



13. Fatty acids are important components of many lipids. For which of the following lipid classes or lipid derivatives are fatty acids **not a significant component**?

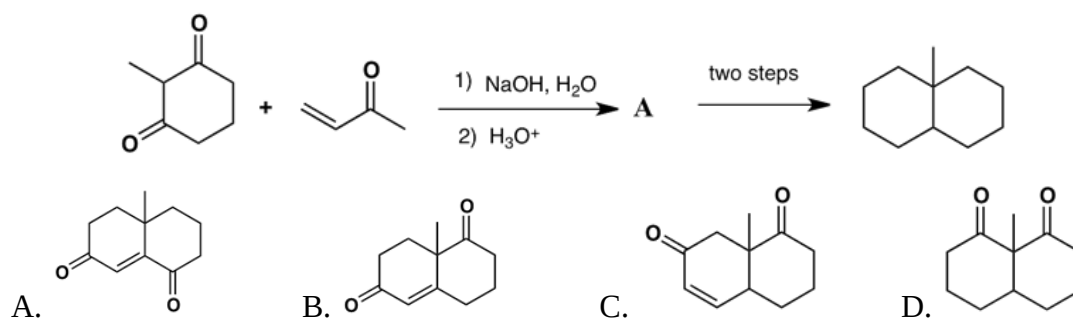
- | | |
|------------------|-------------|
| A. Phospholipids | C. Waxes |
| B. Triglycerides | D. Steroids |

14. Which among the following compounds do not undergo aldol condensation reaction?

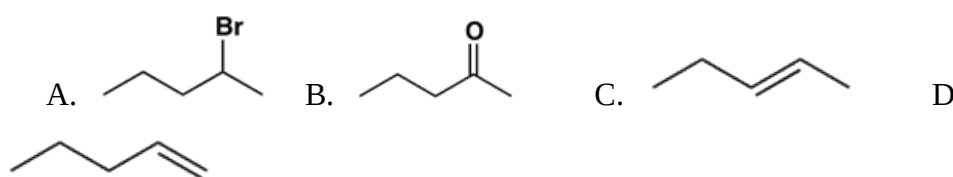
2-Phenylethanal (1), 2,2-Dichloropropanal (2), 2-Bromobenzaldehyde (3), Propanal (4), acetophenone (5)

- | | |
|---------------|---------------|
| A. 1 and 2 | C. 2 and 3 |
| B. 1, 2 and 3 | D. 2, 3 and 4 |

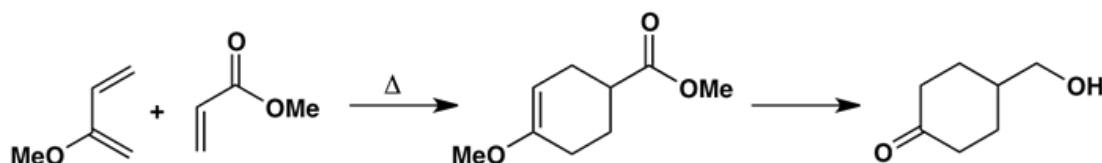
15. Among the compounds listed from (a)-(d), which one of them **matches** with intermediate A formed during the reaction scheme indicated here below?



16. Which of the following is **not** a good alternative target for the synthesis of pentan-2-ol?



17. 4-Hydroxymethylcyclohexanone can be synthesized from a Diels-Alder adduct following the reaction outlined below. Which combination of reagents is appropriate?



- | | |
|--|--|
| A. 1) $\text{NaBH}_4/\text{MeOH}$; 2) H_3O^+ | B. 1) NaBH_4/THF ; 2) $\text{NaOH}/\text{H}_2\text{O}$ |
| C. 1) $\text{LiAlH}_4/\text{Et}_2\text{O}$; 2) H_3O^+ | D. 1) $\text{LiAlH}_4/\text{Et}_2\text{O}$; 2) $\text{NaOH}/\text{H}_2\text{O}$ |

18. For an isothermal Process

A. $W = -Q$.

C. $\Delta U = W + Q$.

B. $\Delta U = W$.

D. $\Delta U = Q$.

19. The entropy of the Universe during the Joule's free expansion will be

A. Always negative

C. Zero

B. Always positive

D. All

20. Select the correct Maxwell's relation from the following

A. $\left(\frac{\partial S}{\partial v}\right)_P = -\left(\frac{\partial P}{\partial T}\right)_S$

C. $\left(\frac{\partial S}{\partial P}\right)_V = \left(\frac{\partial V}{\partial T}\right)_S$

B. $\left(\frac{\partial S}{\partial P}\right)_V = \left(\frac{\partial P}{\partial T}\right)_S$

D. $\left(\frac{\partial S}{\partial v}\right)_P = \left(\frac{\partial P}{\partial T}\right)_S$

21. Maximum number of phases that can be simultaneously exist in equilibrium for a one component system is

A. 1 B. 2 C. 3 D. 4

22. If an ideal gas is allowed to expand adiabatically, the work done by the gas is equal to :

A. The loss of internal energy.

C. The decrease in pressure.

B. The rise in temperature.

D. The loss of entropy.

23. If two systems are separately in thermal equilibrium with the third system then they themselves are in the thermal equilibrium with each other. This is_____

E. First law of thermodynamics

G. Third law of thermodynamics

F. Second law of thermodynamics

H. Zeroth law of thermodynamics

24. A Carnot engine operates between 200°C and 20°C. Its maximum possible efficiency(η) is:

I. 90% B) 100% C) 38% D) 72%

25. If equivalent conductance of NaCl, HCl and C_2H_5COONa at infinite dilution are 126.45, 426.16 and 91 $\text{ohm}^{-1} \text{cm}^2$, respectively, the equivalent conductance of C_2H_5COOH is:

A. 390.71 $\text{ohm}^{-1} \text{cm}^2$

C. 643.61 $\text{ohm}^{-1} \text{cm}^2$

B. 461.61 $\text{ohm}^{-1} \text{cm}^2$

D. 195.355 $\text{ohm}^{-1} \text{cm}^2$

26. Which of the following is an expression at equilibrium for electrochemical cell?

A. $nFE^\circ_{\text{cell}} = RT \ln K$

C. $\Delta G = \Delta G^\circ + RT \ln K$

B. $-nFE^\circ_{\text{cell}} = RT \ln K$

D. $-\Delta G = \Delta G^\circ + RT \ln K$

27. Which of the following is correct?

- A. $\Delta G^\circ < 0$, $K > 0$, and $E^\circ_{\text{cell}} > 0$ for spontaneous cell reaction
- B. $\Delta G^\circ > 0$, $K < 0$, and $E^\circ_{\text{cell}} < 0$ for spontaneous cell reaction
- C. $\Delta G^\circ = 0$, $K = 0$, and $E^\circ_{\text{cell}} = 0$ for spontaneous cell reaction
- D. $\Delta G^\circ < 0$, $K < 0$, and $E^\circ_{\text{cell}} < 0$ for spontaneous cell reaction
28. The value of E°_{cell} for the following reaction is 1.260 V. What is the value of E_{cell} given the concentrations as shown? $2\text{Al(s)} + 3\text{Cd}^{2+}(\text{aq}, 0.1 \text{ M}) \rightarrow 3\text{Cd(s)} + 2\text{Al}^{3+}(\text{aq}, 0.6 \text{ M})$
 $E^\circ_{\text{cell}} = 1.260 \text{ V}$
- A. 1.235 V
- B. 1.285 V
- C. 1.150 V
- D. 1.370 V
29. Which of the following is NOT correct according to The Kinetic Molecular Theory of Gases?
- A. Gases consist of a large number of tiny particles (atoms and molecules)
- B. The collisions between the molecules and the walls are perfectly inelastic
- C. The average kinetic energy of the gas particles changes with temperature
- D. The molecules do not exert any force of attraction or repulsion on one another
30. Based on the following experimental data for reaction $2\text{NO(g)} + 2\text{H}_2 \rightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O(g)}$, the rate law expression is

Experiment	[NO]	[H ₂]	Rate (M/s)
1	0.005	0.002	1.25×10^{-5}
2	0.010	0.002	5×10^{-5}
3	0.010	0.004	1×10^{-4}

- A. Rate = $k[\text{NO}][\text{H}_2]$
- B. Rate = $k[\text{NO}]^2[\text{H}_2]$
- C. Rate = $k[\text{NO}][\text{H}_2]^2$
- D. Rate = $k[\text{NO}]^4[\text{H}_2]^2$
31. In a laboratory experiment 50.00 cm³ of a solution originally 0.2 molar in CH₃COOH was permitted to come in equilibrium with a 5.00-gram sample of charcoal. At equilibrium, 25.00 cm³ of the CH₃COOH solution required 30.00 cm³ of 0.10 N NaOH for neutralization. Calculate the weight of CH₃COOH adsorbed per gram of charcoal and the equilibrium concentration of the acid.
- A. 0.048 gm
- B. 0.48 gm
- C. 4.8 gm
- D. 0.0048 gm
32. Equilibrium:

- A. Is defined as equal concentration of reactants and products
 - B. Is defined as equal rates for forward and reverse reactions
 - C. Can be shifted by adding a catalyst
 - D. Can exist for chemical changes but not for physical changes
- 33.** Which one of the following is true about electrolytic conductance?
- A. Temperature increases, conductance also increases.
 - B. Higher dielectric constant shows lower conductance in solution.
 - C. Molar conductance increases with increase in dilution.
 - D. Conductivity increases with decrease in viscosity.
- 34.** Which of the following statements is true about metallic conductors?
- A. Occurs due to the flow of ions
 - B. Occurs due to the flow of electrons
 - C. Occurs due to the diffusion of solvents
 - D. Occurs due to the flow of solutes
- 35.** Eutrophication is
- A. The dissolution of oxygen in water bodies to a level that supports aquatic life
 - B. The depletion of oxygen from water bodies due to excessive growth of algae
 - C. The phenomena caused by excessive introduction of sulphate and carbon dioxide to water bodies.
 - D. The production of oxygen up on photosynthesis
- 36.** What are the permissible limits of the water quality parameters pH, FC, and TDS for drinking water in Ethiopia?
- | | |
|----------------------------------|--------------------------------------|
| A. 5.5-7.5, 100 /100ml, 500 mg/l | B. 6.5-8.5, 0 /100ml, <100 mg/l |
| C. 6-9, 0 /100ml, 1500 mg/l | D. 6.5-8.5, 0 unit/100ml, <2000 mg/l |
- 37.** Which gas in the atmosphere most contributes to photochemical smog?
- A. SO₂ B. CO C. NO_x D. CO₂
- 38.** What does the BOD₅ test measure in a wastewater treatment plant?
- A. The concentration of organic material decomposed by aerobic and anerobic bacteria in a waste sample.
 - B. The concentration of organic material and reduced forms of nitrogen (e.g., ammonia) in a waste sample.
 - C. The oxygen-depleting capacity of a waste sample.
 - D. The dissolved-oxygen concentration of a waste sample.

39. The ions formed by dissolution of minerals in water and/or added from anthropogenic sources, such as Ca^{2+} , HCO_3^- , NO_3^- , and PO_4^{2-} control the chemical conditions of an aquatic system. Which of the following aquatic chemical processes not that much influence the concentration, transport and fate of these major species?
- A. acid-base reaction B. complexation reaction
C. oxidation-reduction reaction D. precipitation-dissolution reaction
40. Which of the following chemical cycle includes the process photosynthesis, dissolution of HCO_3^- , chemical precipitation of CO_2 and processing of petroleum product?
- A. Nitrogen cycle C. Carbon cycle
B. The phosphorus cycle D. the water cycle
41. The potential of soil to store plant nutrients and toxic chemicals in soil humus depend on its
- A. surface charge B. pH C. Surface area D. All
42. In which of the following process no mass cross across the system boundaries till the process completion?
- A. Batch process B. Continuous process C. Semi-batch process D. Semi-continuous
43. The type of water treatment method which involves forcing pure water through a semipermeable membrane that allows the passage of water but not of other materials?
- A. Chlorination B. Electro dialysis C. Reverse osmosis D. Ion-exchange
44. The water treatment process which is used mostly to remove colloidal impurity is:
- A. Plain sedimentation B. Screening C. Filtration D. Coagulation
45. One of the following is not derivative of nitrogen?
- A. Ammonia B. Nitric acid C. TNT D. Glycerol
46. Which one of the following is not a double fertilizer?
- A. DAP B. Urea C. MAP D. Nitro phosphate
47. Identify the process which is not involved in the manufacturing of urea from the given alternatives?
- A. Drying B. Decomposition C. Carbamate formation D. All
48. One of the following fabrication methods of polymers is used for fabrication of thermoplastic bottles?
- A. Blow molding B. Compression molding C. Injection molding D. All
49. One of the following is a process of introducing oil into the skin before the leather is dried to replace the natural oils lost in the beam house and Tanyard processes?
- A. Pickling B. Trimming C. Fat liquoring D. Sweltering

50. Which of the following process in beer production in the preparation of wort which involves mixing of the flour of malt with water and subjected to certain processes to obtain wort with suitable composition?
- A. Stabilization B. Maturation C. Milling D. Mashing
51. Which one of the following in coal processing technique used to convert solid coal into liquid fuel?
- A. Hydrogenation B. Gasification C. Carbonization D. Cracking
52. One of the following is false about condensation polymerization.
- A. Produce low molecular weight polymers
B. Produce small molecule byproducts
C. Involves initiation, propagation, termination
D. Requires a di-functional monomer
53. A process of mixing malt flour with water and hops followed by boiling at elevated temperature to prepare wort in beer industry.
- A. Malting B. Mashing C. Fermentation D. Brewing
54. In atomic absorption spectroscopy, which of the following is the generally used radiation source?
- A. Tungsten lamp
B. Hydrogen or deuterium discharge lamp
C. Hollow cathode lamp
D. Xenon mercury arc lamp
55. What happens during the 'elution from the column' phase in chromatography?
- A. Components with greatest affinity elute first
B. Components elute in a random manner
C. Components with least affinity elute first
D. Components elute according to their concentration in the mixture
56. The unit molarity (mole per liter) is used to express the concentration of a certain species of chemical element in a concentrated solution. Meanwhile, other more appropriate units are more commonly used to describe the concentration of more diluted solutions. One of these units, which is very close to the milligram per liter measurement, is?
- A. Part per million (ppm) C. Part per thousand (ppt)
B. Part per trillion (ppt) D. Part per billion (ppb)
57. In chemical equilibrium, a principle states that if a stress (for example, a change in concentration, pressure, temperature or volume of the vessel) is applied to a system in

equilibrium, the equilibrium will shift in such a way to lessen the effect of the stress. This principle is called?

- A. The Hess Principle
- B. The Le Chatelier's Principle
- C. The Aufbau Principle
- D. The Robert Principle

58. In complexometric titration, the most common used chelating agent is EDTA (ethylenediamine tetraacetic acid). EDTA's molecules will combine with metals to form chelate. EDTA is classified as a?

- A. Tetradentate ligand
- B. Hexadentate ligand
- C. Bidentate ligand
- D. Monodentate ligand

59. All of the following methods are used for the analysis of precipitate, except?

- A. Mohr's method
- B. Fajan's method
- C. Volhard's method
- D. Haber's method

60. As suggested by its name, an infrared spectrometer uses infrared electromagnetic radiation in its machine. This infrared radiation will result in which molecular activity?

- A. Molecular rotation
- B. Molecular ionization
- C. Molecular vibration
- D. Molecular movement

61. A typical infrared (IR) spectrum is divided into 2 parts, namely the left part with the absorption peaks at $1400 - 4000 \text{ cm}^{-1}$ and the right part with the absorption peaks at $600 - 1400 \text{ cm}^{-1}$. Beginners in this field only study the left part. The right part is studied by expert spectroscopists. What is the name given to this right part?

- A. Stretching region
- B. Complex splitting region
- C. Unsymmetrical region
- D. Fingerprint region

62. In which of the following ways, absorption is related to transmittance?

- A. Absorption is a multiple of transmittance
- B. Absorption is the negative logarithm of transmittance
- C. Absorption is the logarithm of transmittance
- D. Absorption is the reciprocal of transmittance

63. Chromatography is a physical method that is used to separate and analyses _____

- A. Viscous mixtures
- B. Complex mixtures
- C. Simple mixtures
- D. Metals

64. Which of the following types of chromatography involves the process, where the mobile phase moves through the stationary phase by the influence of gravity or capillary action?

- A. High pressure liquid chromatography
- B. Planar chromatography

- C. Column chromatography
 - D. Gas chromatography
65. In-gel permeation chromatography, _____ is eluted first from the column
- A. Smaller molecules
 - B. Bigger molecules
 - C. Cations
 - D. Anions
66. Proteins may be separated according to size by
- A. Reverse-phase chromatography
 - B. Isoelectric focusing
 - C. Molecular exclusion chromatography
 - D. Ion-exchange chromatography
67. Transmittance is given as $T = P/P_o$. If P_o is the power incident on the sample, what does P represent?
- A. Radiant power transmitted by the sample
 - B. Radiant power absorbed by the sample
 - C. Sum of powers absorbed and scattered
 - D. Sum of powers transmitted and reflected
68. Absorption of radiation in the UV range attributable to $n \rightarrow \pi^*$ electronic transitions is characteristic of which of the following types of compounds?
- A. Conjugated polyenes
 - B. Aromatic hydrocarbons
 - C. Non-conjugated polyenes
 - D. Unsaturated carbonyl compounds
69. Compound A has greater shielding constant than compound B. Which of them will have more chemical shift?
- A. Both will have equal chemical shifts
 - B. Compound A
 - C. Compound B
 - D. Chemical shift has no relation with shielding constants
70. NMR is the study of absorption of _____ by nuclei in a magnetic field?
- A. Radio frequency radiation
 - B. Radioactive radiation
 - C. IR radiation
 - D. Microwaves

71. Which of the following are considered to be the lowest energy form of electromagnetic radiation?
- A. Radio waves
 - B. IR radiation
 - C. Micro waves
 - D. UV radiation
72. The amount of energy available in radio frequency radiation is sufficient for which of the following?
- A. Affect the nuclear spin of an atom
 - B. Excite an atom
 - C. Vibrate an atom
 - D. Vibrate a molecule
73. Spectroscopy measures the change in behavior of a molecule when it is exposed to which of the following?
- A. An electrical charge
 - B. A centrifugal force
 - C. Electromagnetic radiation
 - D. Acidic conditions
74. In size exclusion chromatography, solute molecules are separated based on ____
- A. Molecular phase
 - B. Molecular geometry and size
 - C. Molecular composition
 - D. Molecular formula
75. In a chromatographic separation, which of the following is most appropriate for the qualitative analysis of a substance?
- A. Capacity factor
 - B. Retention time
 - C. Taking factor
 - D. Resolution
76. The basis of the technique of chromatography for separating components of a mixture is?
- A. The absorption of infrared radiation by the components.

- B. The interaction of the components with a stationary and a mobile phases
 - C. The differing movement of particles of different mass in an electrical field
 - D. the deflection of charged particles in a magnetic field
77. The correct statement from the following is _____
- A. If the polarity of solute is greater, it will be adsorbed on a polar surface much weekly
 - B. If the polarity of solute is less, it will be adsorbed on a polar surface much strong
 - C. If the polarity of solute is greater, it will be adsorbed on a polar surface much strongly
 - D. All
78. In chromatogram, the position of peaks on the time axis can be used to determine which of the following?
- A. Amount of component in the sample
 - B. Components of the sample
 - C. Column efficiency
 - D. Column resolution
79. Voltammetry corresponds to measurement of _____
- A. Current
 - B. Voltage
 - C. Potential
 - D. concentration
80. Identification of the reducible species can be done by knowing
- A. Diffusion current
 - B. Half wave potential
 - C. Discharge potential
 - D. All
81. In polarography, wave height is equal to
- A. Limiting current
 - B. Diffusion current
 - C. Decomposition potential
 - D. All
82. Which of the following quantum numbers governs the spatial orientation of an atomic orbital?
- A. Spin quantum number
 - B. Azimuthal quantum number
 - C. Principal quantum number
 - D. Magnetic quantum number
83. An element with electronic configuration $[X]ns^2(n-1)d^{10}$ belongs:
- A. Noble gas
 - B. Alkaline element
 - C. Transition element
 - D. s-block element
84. The geometry of unknown compound XY_3 in which X and Y are group VIIA elements is:
- A. Trigonal bipyramidal
 - B. Octahedral
 - C. Square planar
 - D. None
85. Which one of the following statements is incorrect about periodic characters of an element?

- A. Elements to the left have pronounced metallic character while to the right nonmetallic character.
 - B. Electro-positivity is a tendency of an element to gain electrons and hence decrease down a group.
 - C. Elements to the bottom have lower electron affinity since increment of atomic size.
 - D. Elements to the left have lower ionization energy due to decrease in nuclear charges.
- 86.** The type of hydride that many transition metals form, in which H_2 molecules and H atoms occupy the holes in the crystal structure of the metal, is _____.
- A. Covalent hydride B. Network hydride C. Interstitial hydride D. ionic hydride
- 87.** Which statement describes the reactivity of Alkaline earth metals?
- A. The metals reduce O_2 to form the oxide
 - B. Members of this group reacts with halogen to form ionic halides
 - C. Most of the metals reacts with nitrogen to form ionic nitrides
 - D. All of the above are correct
- 88.** What is the point group of linear molecules with a center of symmetry?
- A. $D_{\infty h}$ B. $C_{\infty h}$ C. T_d D. None of these
- 89.** Which of the following is not true?
- A. Charge transfer (CT) gives rise to intense absorptions
 - B. 'd-d' absorption bands are weak
 - C. Charge transfer (CT) absorptions often occur at higher energies
 - D. ligand-centered $n-\pi^*$ and $\pi-\pi^*$ bands often occur at higher energies
- 90.** The exclusion rule states that, if a molecule has a center of inversion,
- A. all of its modes can be both IR and Raman active
 - B. some of its modes can be both IR and Raman active
 - C. none of its modes can be both IR and Raman active
 - D. A and B
- 91.** Which vibrational mode of CO_2 is IR inactive?
- A. Symmetric stretch C. Bend (deformation)
 - B. Asymmetric stretch D. All
- 92.** The vibrational modes of F_2O molecule occur at 928 , 831 and 461 cm^{-1} are all IR active. To what point group does F_2O belong?

A. D_{2h} B. $C_{\infty h}$ C. $D_{\infty h}$ D. C_{2v}

93. Which of the following is not the symmetry element of H_2 molecule?

A. i B. σ_h C. $2C_{\infty}$ D. None of these

94. Which compound(s) is/are responsible for the formation of a voluminous product while heating $(NH_4)_2Cr_2O_7$

A. NH_3 B. H_2O C. Cr_2O_3 D. A and B

95. What is the color of the product observed during heating $(NH_4)_2Cr_2O_7$?

A. Orange B. yellow C. green D. red

96. Which ion or compound will be produced when an aqueous solution of $KMnO_4$ treated under acidic conditions?

A. MnO_2 B. Mn^{2+} C. MnO_4^- D. $MnO(OH)_2$

97. The correct order of the stoichiometries of $AgCl$ formed when $AgNO_3$ in excess is treated with complexes: $CoCl_3.6NH_3$, $CoCl_3.5NH_3$, $CoCl_3.4NH_3$ respectively is

A. $3AgCl, 1AgCl, 2AgCl$

B. $3AgCl, 2AgCl, 1AgCl$

C. $2AgCl, 3AgCl, 2AgCl$

D. $1AgCl, 3AgCl, 2AgCl$

98. Mohr's salt is

A. $FeSO_4.(NH_4)_2SO_4.6H_2O$

C. $MgSO_4.7H_2O$

B. $Fe_2(SO_4)_3.(NH_4)_2SO_4.6H_2O$

D. $FeSO_4.7H_2O$

99. Acetylacetonate is a common _____ that used for the synthesis of different metal complexes

A. Monodentate ligand

C. Bidentate ligand

B. Tri dentate ligand

D. Tetra dentate ligand

100. What is not true about $SnCl_2$ and $SnCl_4$?

A. $SnCl_2$ solid while $SnCl_4$ Liquid at room temperature

B. $SnCl_4$ is Covalent and $SnCl_2$ is ionic

C. $SnCl_2$ is more stable than $snCl_4$

D. $SnCl_2$ is a good oxidizing agent while $SnCl_4$ reducing

